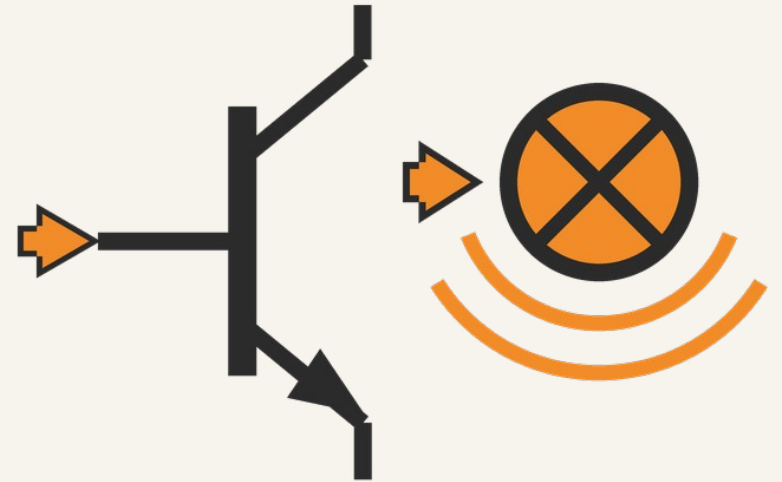


BOOKLET 3 • ELECTRONICS AND ANALOGUE
CONTROL

Transistors and Relays

A small voltage at the base, switching
something much bigger.

National 5 Engineering Science



Daily Review

Lesson 5 — Voltage Dividers and Sensors

1 Which resistor is V_2 (V_{in}) taken across?

the lower one, R_2

2 An LDR is R_2 and it gets darker. What happens to V_2 ?

V_2 rises

3 $R_1 = 3 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $V_1 = 6 \text{ V}$. Find V_2 .

$V_2 = 2 \text{ V}$

4 Name an output transducer that produces movement.

a motor

5 Write the full name of an LDR.

light dependent resistor

PART 1

The Transistor as a Switch

The divider gives you a changing voltage. How does that turn a lamp on?

By the end of this part...

- 1 I can describe the transistor as a switching device
- 2 I can describe how a small input voltage switches a larger output

What a transistor does

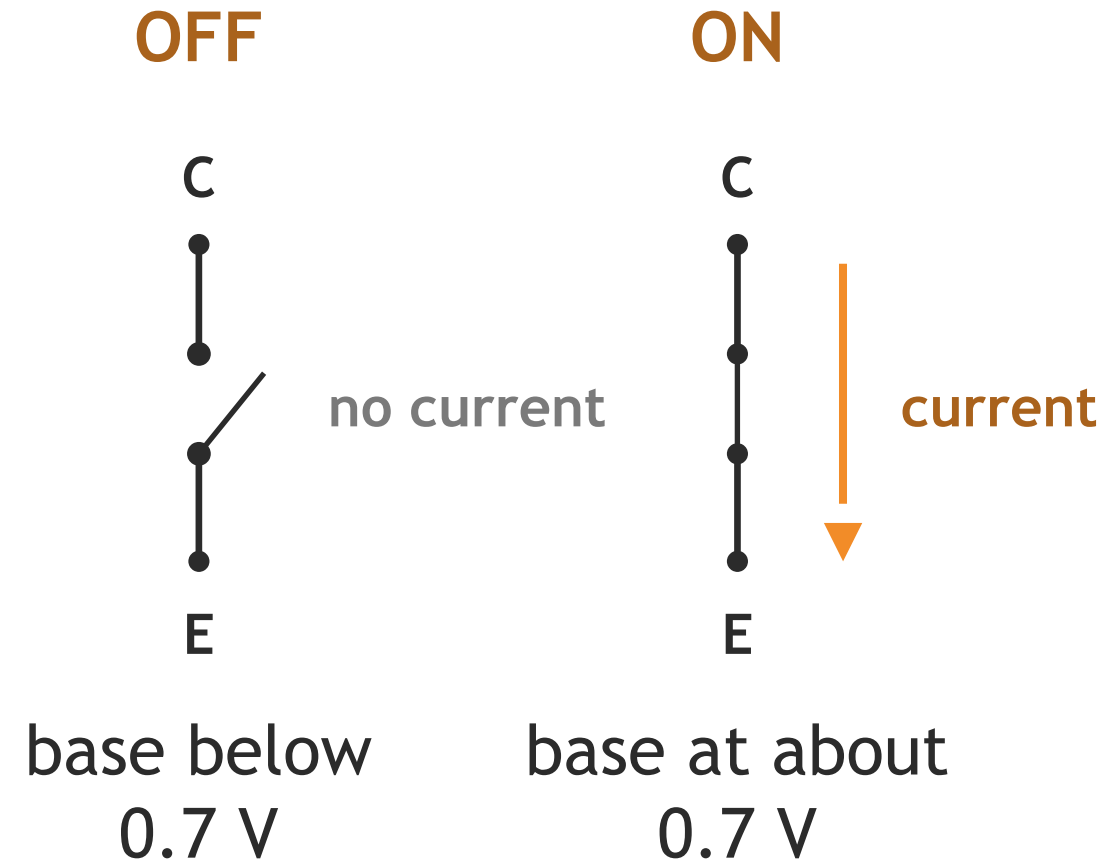
Booklet p.27

An electronic switch

A small voltage at the base switches on a much larger current from collector to emitter.

At National 5

You only need the NPN transistor.



The NPN symbol and its connections

Booklet p.27 – the symbol on the data sheet

B – base

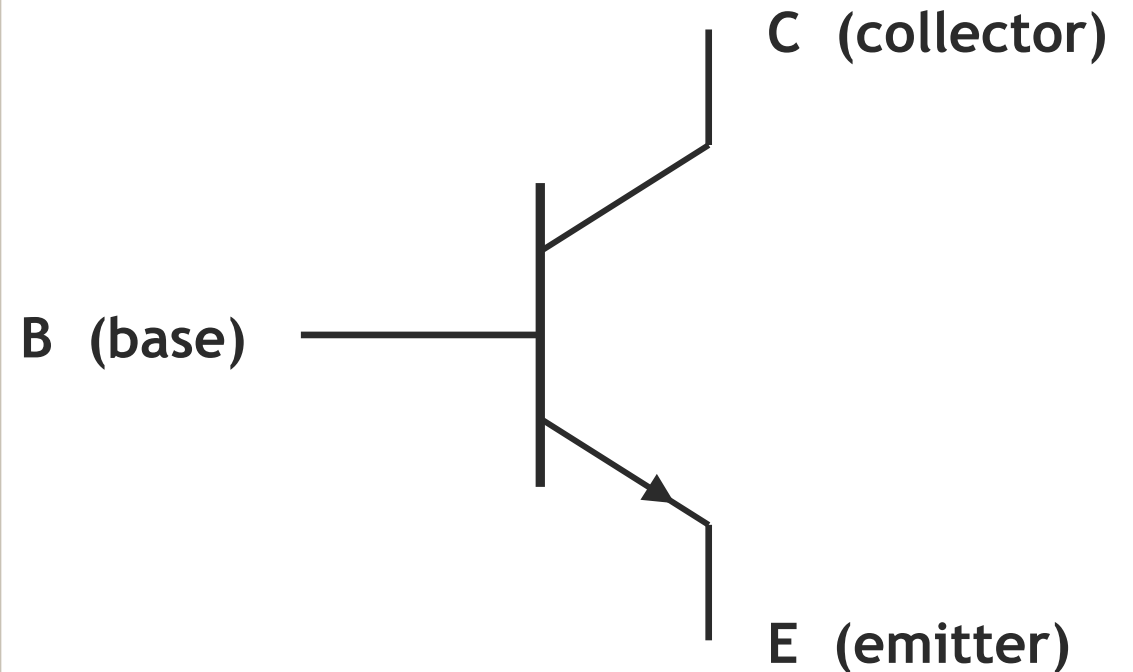
the input: a small voltage
here switches it

C – collector

connected to the output
device

E – emitter

the path back to 0 V



Real transistors

One symbol, many cases

Different cases

Transistors come in many shapes and sizes. The symbol is the same for all of them.

At your bench

Find B, C and E from the component data sheet (rapid)

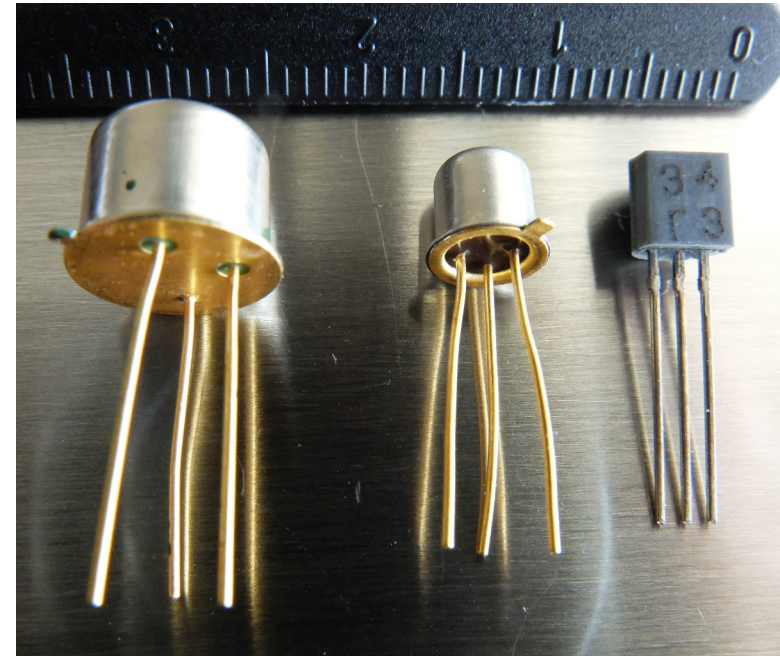


Photo: Drahtlos,
CC BY-SA 4.0, Wikimedia Commons

The switching voltage

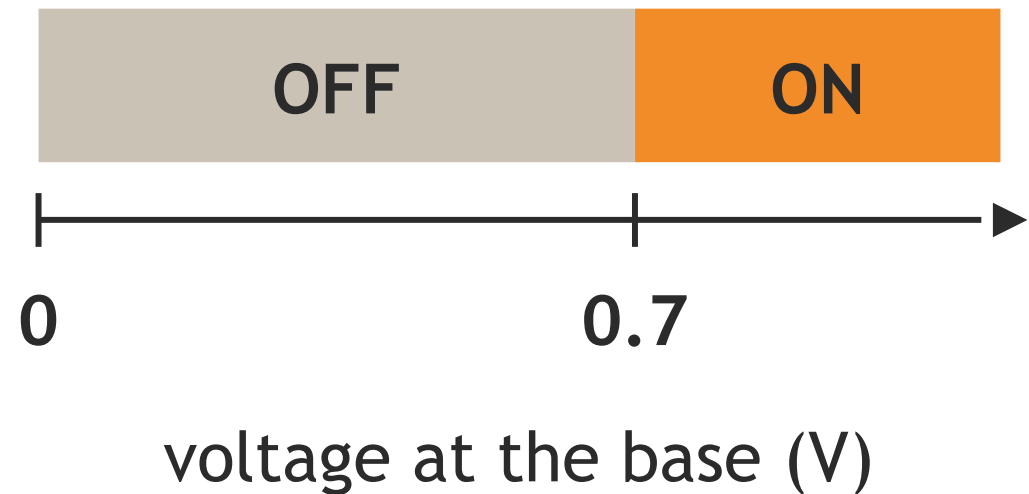
Booklet p.27

Below about 0.7 V

at the base, the transistor is OFF.

At about 0.7 V

it switches ON, and the output device operates.



The base resistor

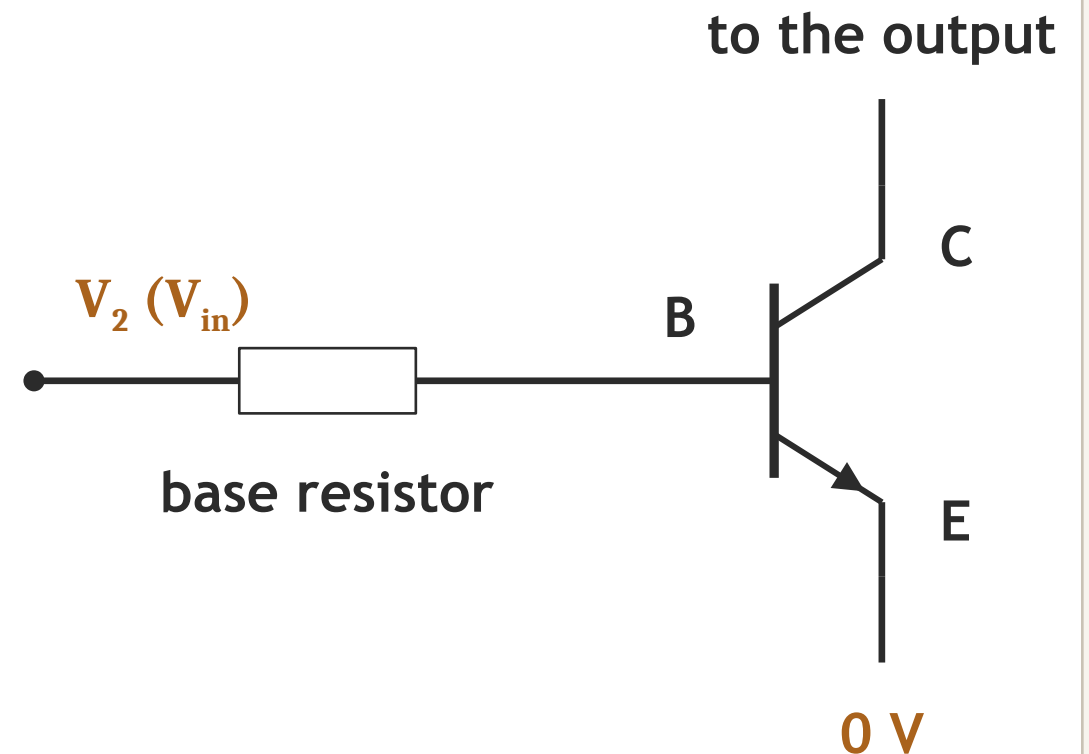
Between the voltage divider and the base

Where it goes

Between the divider's output, V_2 (V_{in}), and the base.

What it does

It limits the base current, so the transistor is not damaged.



A transistor switching a lamp

Booklet p.27

The divider

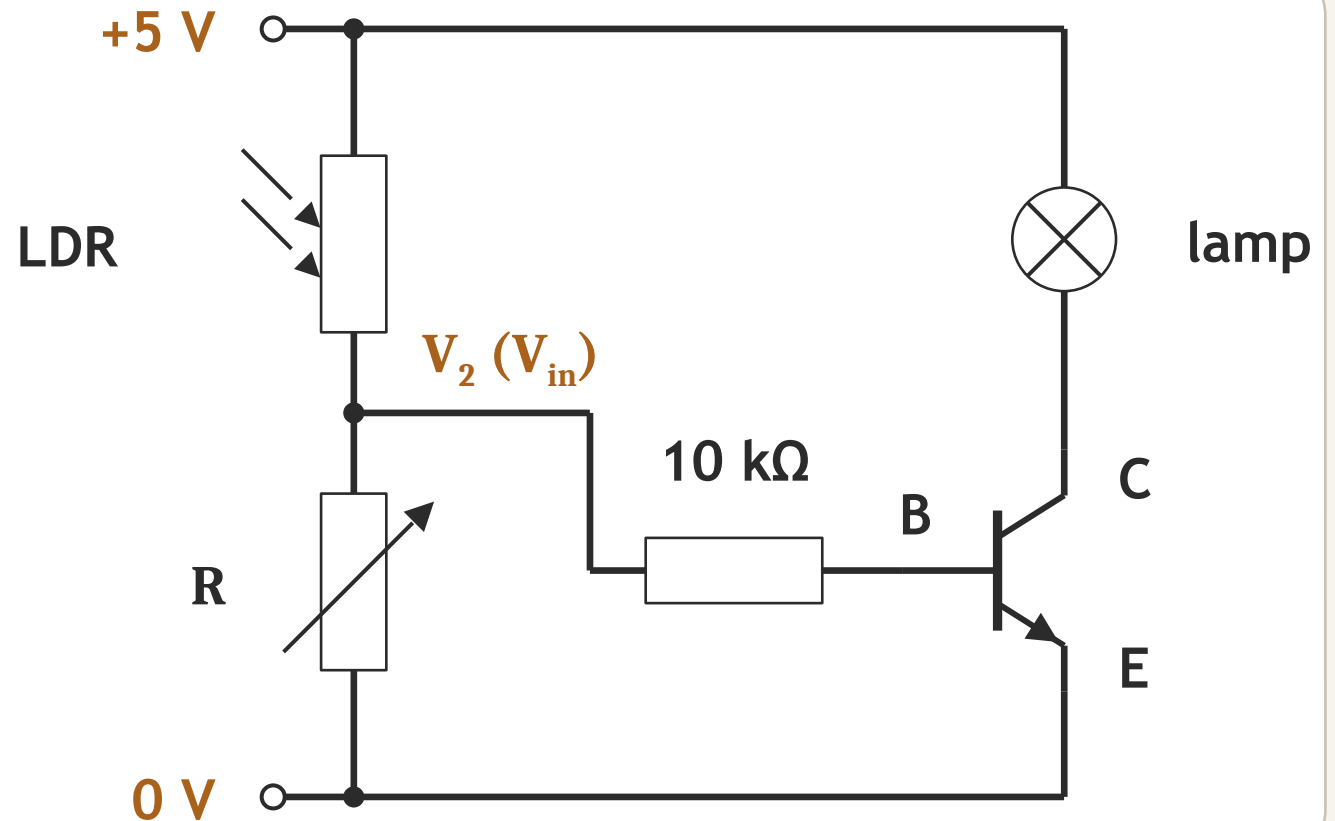
sets V_2 (V_{in}).

The base resistor

limits the base current.

The transistor

switches the lamp on at about 0.7 V.



Show me – the transistor

SHOW ME BOARDS

1

Name the three connections of an NPN transistor.

base, collector, emitter

2

Which connection is the input?

the base

3

What is the approximate switching voltage?

about 0.7 V

4

The base is at 0.3 V. Is the transistor on or off?

off

5

What does the base resistor protect?

the transistor

Show me – the transistor

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Which connection goes to 0 V?
- 2 Which connection goes to the output device?
- 3 The base is at 0.9 V. Is the transistor on or off?
- 4 The base reaches 0.7 V. What does the output device do?
- 5 What sets the voltage at the base in a sensor circuit?

the emitter

the collector

on

it switches on

the voltage divider

Checkpoint

1

Name the component between the divider and the base.

a resistor – the base resistor

2

V_2 (V_{in}) never rises above 0.4 V. What does the output do?

stays off

3

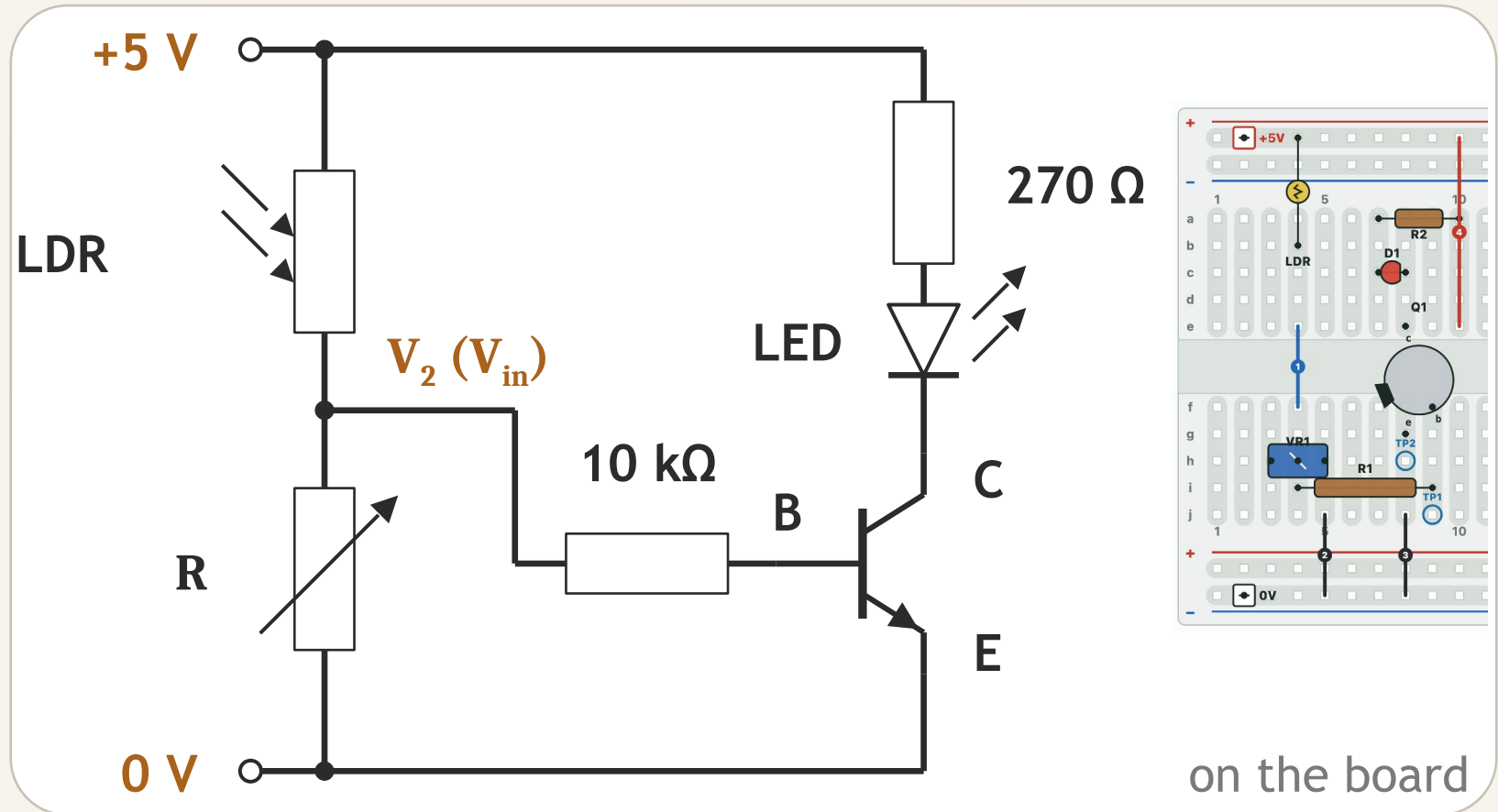
When it is on, the big current flows between which two connections?

collector and emitter

Practical 3 – The Transistor as a Switch

Practical sheet on Teams · success criteria 5.1, 5.2 and 5.6

Find the voltage that switches the LED on. Simulate it in Yenka, then build it.



Daily Review

Part 1 — The Transistor as a Switch

- 1 What is the switching voltage of an NPN transistor?
about 0.7 V
- 2 Which connection is the input?
the base
- 3 What does the base resistor limit?
the base current
- 4 The base is at 0.2 V. Is the output on or off?
off
- 5 What sets the voltage at the base?
the voltage divider

PART 2

Relays and the Protection Diode

How can a 5 V circuit switch a 230 V fan without touching it?

By the end of this part...

- 1** I can describe the function of a relay in a control circuit
- 2** I can describe the function of a protection diode
- 3** I can interpret a full sensor control circuit
- 4** I can simulate a transistor switching circuit with a sensor input

What a relay is

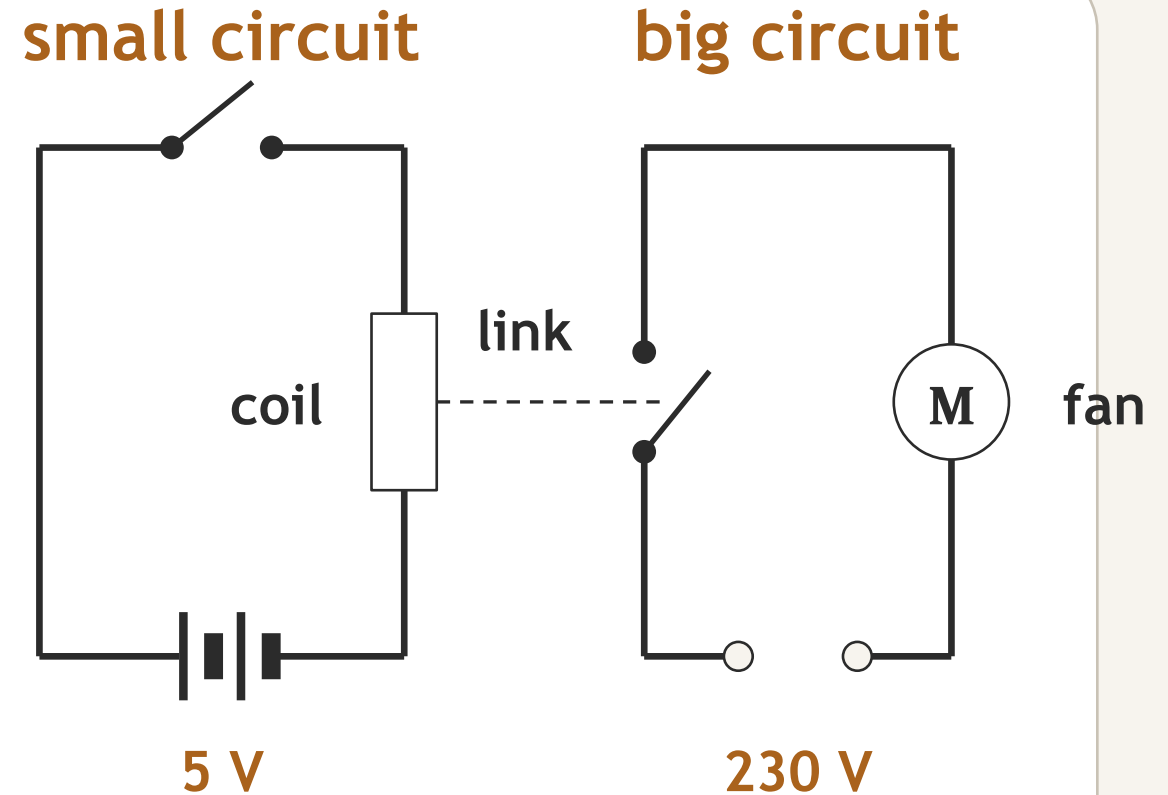
Booklet p.28

An electromagnetic switch

A small current in the coil closes a separate set of contacts.

Two circuits

The coil's circuit and the contacts' circuit are not joined by any wire.



Inside a relay

Booklet p.28

The coil

Current through the coil makes it an electromagnet.

The contacts

The magnet pulls a metal arm across, and the contacts close — you hear a click.

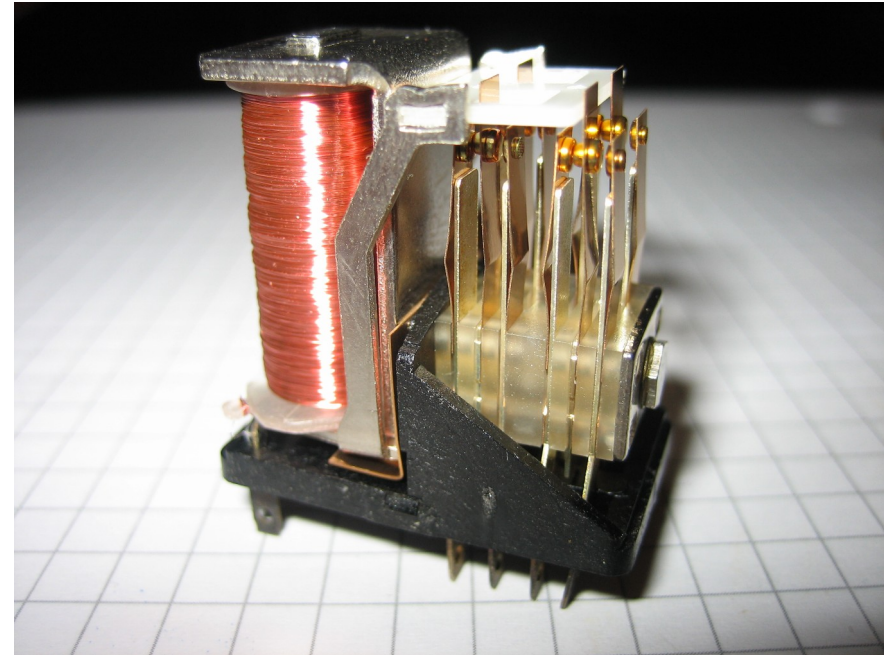


Photo: redhat,
public domain, Wikimedia Commons

Why engineers use a relay

Booklet p.28

The problem

A transistor circuit works at a low voltage. It cannot carry a mains current.

The answer

The small circuit switches the coil, and the contacts switch the big one.



Photo: Dwight Sipler,
CC BY 2.0, Wikimedia Commons

A changeover relay

Two outputs, one relay

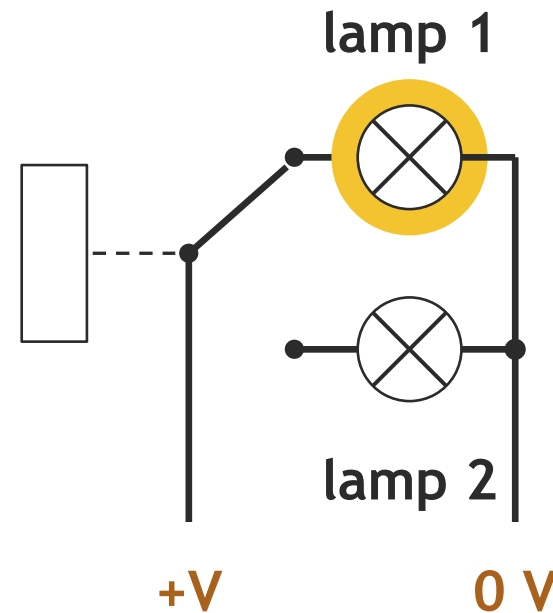
Coil off

The switch arm rests on one contact, so lamp 1 is on.

Coil on

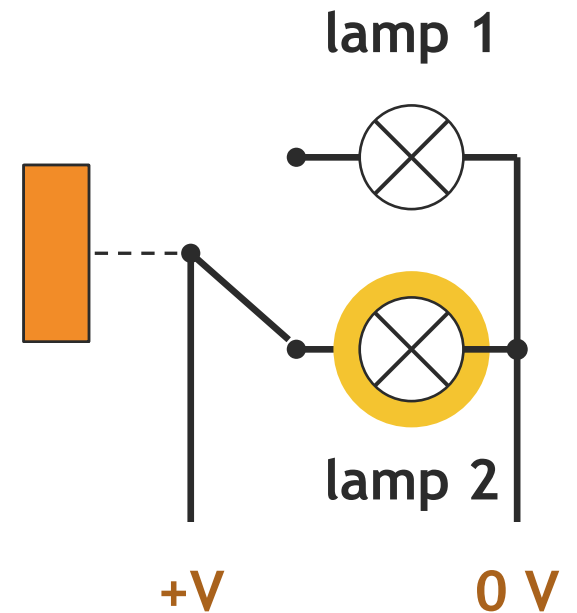
The arm moves to the other contact: lamp 1 goes off and lamp 2 comes on.

coil OFF



lamp 1 on

coil ON



lamp 2 on

The protection diode

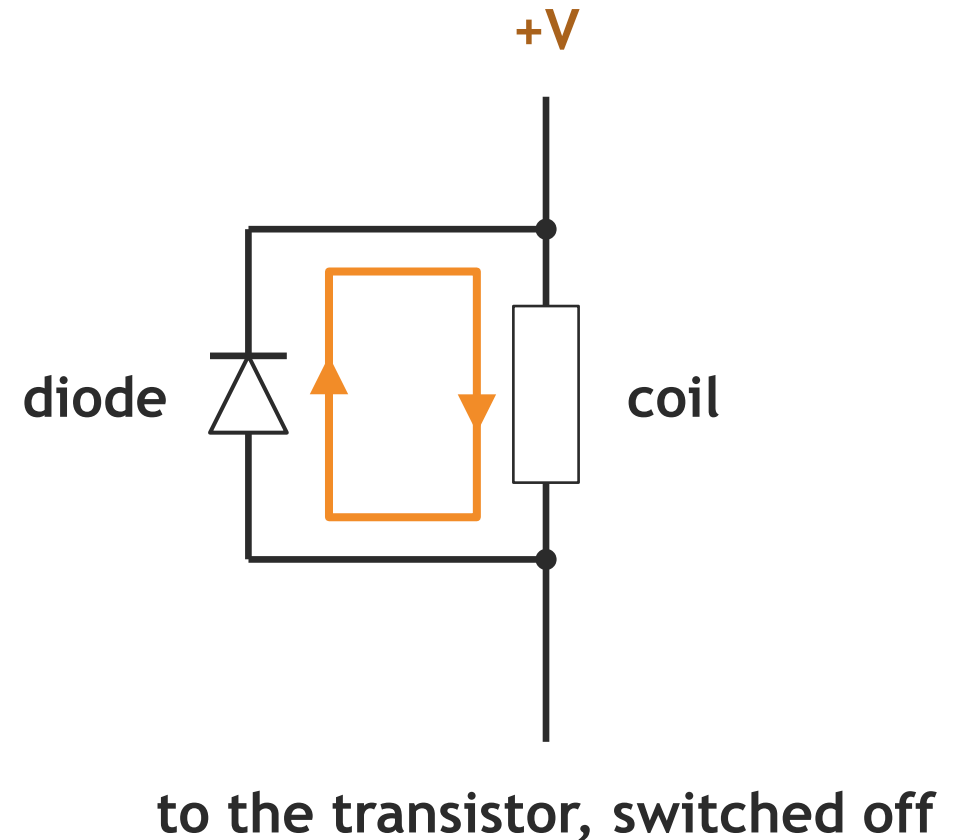
Booklet p.28

The problem

Switching the coil off makes a voltage spike that can damage the transistor.

The answer

A diode across the coil, in reverse to the supply, gives the spike's current a safe path.



Watch out

THE MISTAKE

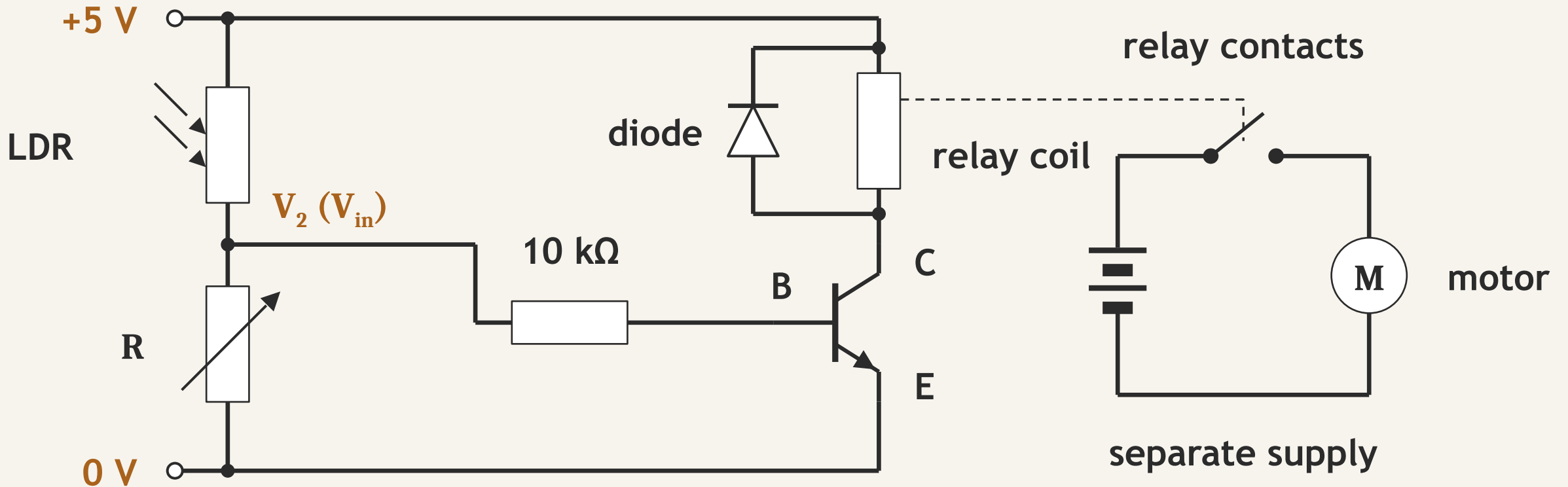
- "A diode limits the base current."
- The diode sits across the coil, not at the base.

THE CORRECT ANSWER

- The base resistor limits the base current.
- The diode protects it from the coil's spike.

The full control circuit

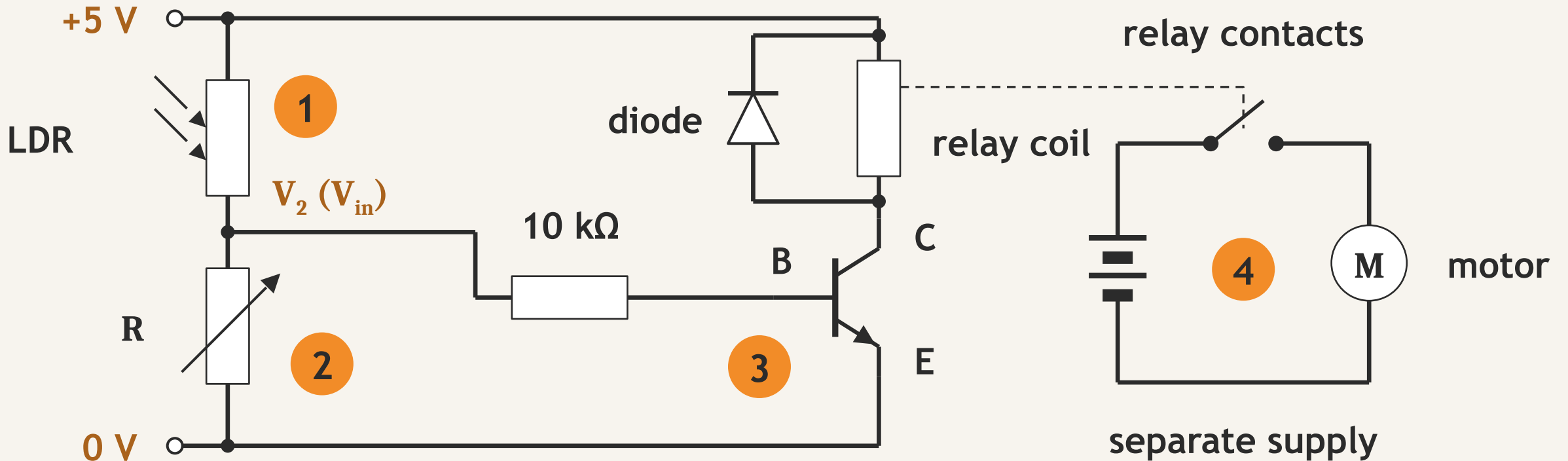
Booklet p.28 – light-sensing



Light-sensing: the LDR is at the top of the divider.

The four blocks of it

Booklet p.28



1 input transducer

2 voltage divider

3 transistor

4 relay and output

Light- and dark-sensing

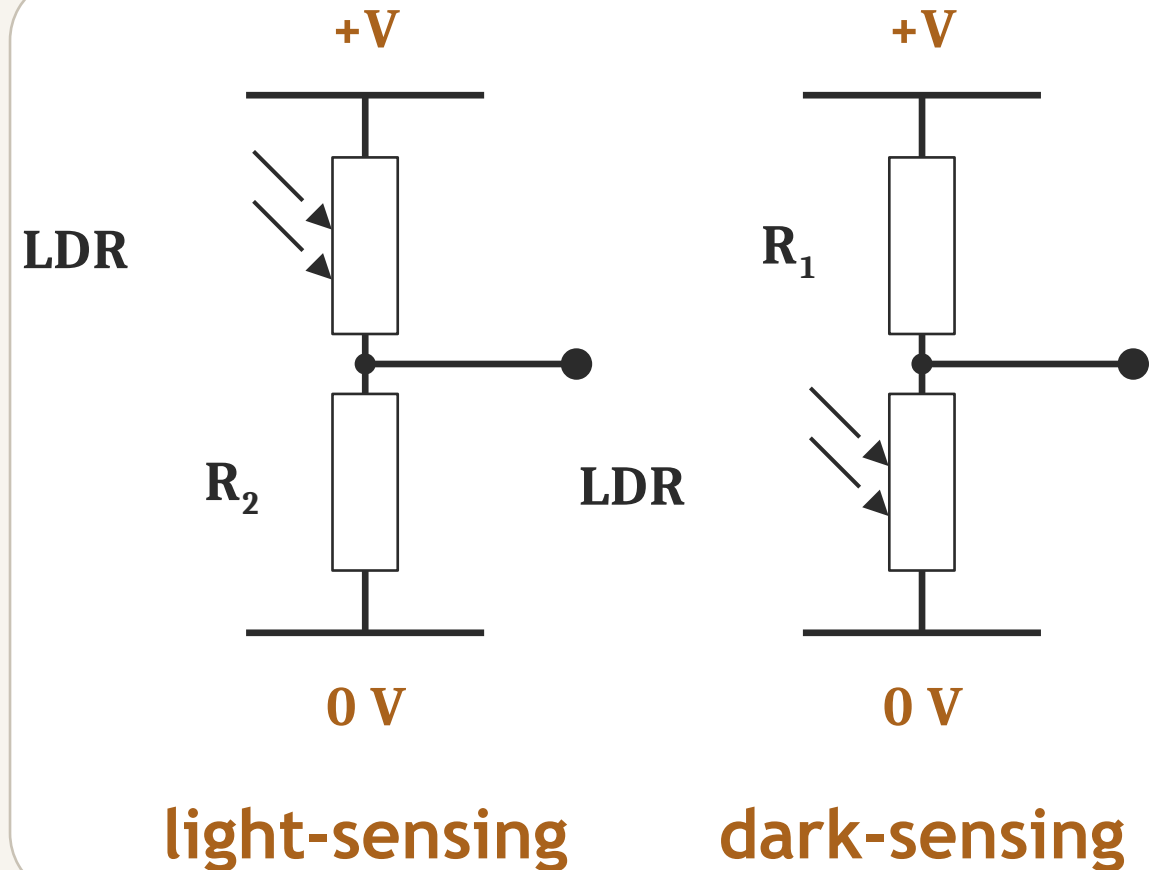
Booklet p.28

Light-sensing

LDR at the top, as R_1 . The output switches on when the light level is high.

Dark-sensing

LDR at the bottom, as R_2 . The output switches on when the light level is low.



Hot and cold sensing

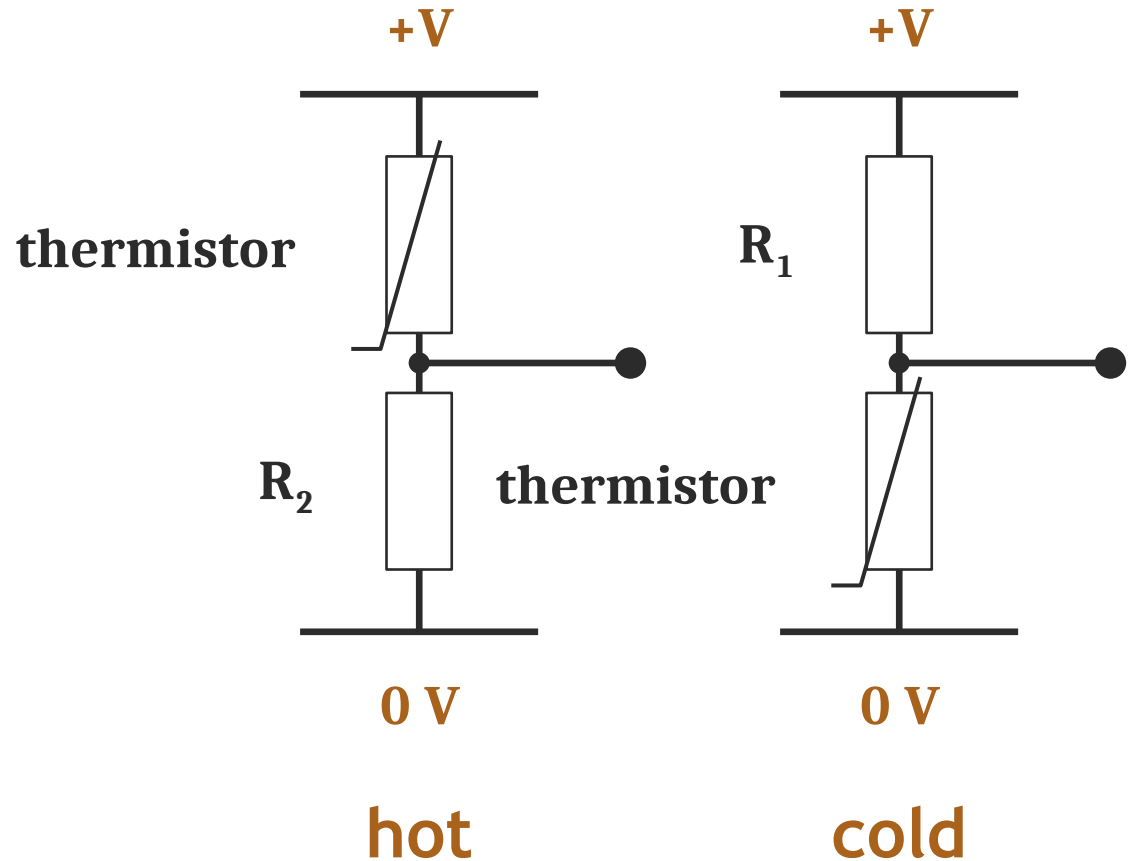
Booklet p.28

High temperature

Thermistor at the top, as R_1 .
The output switches on when it is hot.

Low temperature

Thermistor at the bottom, as R_2 . The output switches on when it is cold.



Show me – relays and sensing

SHOW ME BOARDS

1

What does a relay coil do when current flows through it?

becomes a magnet and closes the contacts

2

Name the component connected across a relay coil.

a (protection) diode

3

V_2 (V_{in}) must rise as it gets colder.
Where does the thermistor go?

at the bottom, as R_2

4

V_2 (V_{in}) must rise as it gets brighter.
Where does the LDR go?

at the top, as R_1

5

What limits the current into the base?

the base resistor

Show me – relays and sensing

SHOW ME BOARDS

EXTRA PRACTICE

1

Is a relay's coil joined to its contacts by a wire?

no – only by magnetism

2

When does a relay coil make a voltage spike?

when it switches off

3

V_2 (V_{in}) must rise as it gets darker.
Where does the LDR go?

at the bottom, as R_2

4

A changeover relay switches on. What happens to its two outputs?

one goes off, the other comes on

5

The protection diode's bar faces which supply rail?

the +V rail

Describing it – the routine

Booklet p.31 – Knowledge Organiser

In this order

- 1 the sensor's resistance
- 2 V_2 (V_{in})
- 3 the transistor
- 4 the relay, then the output

A pipe heater – thermistor as R_2

the temperature falls

the thermistor's resistance rises

V_2 (V_{in}) rises

transistor and relay switch on

the heater switches on

What a 5-mark description looks like

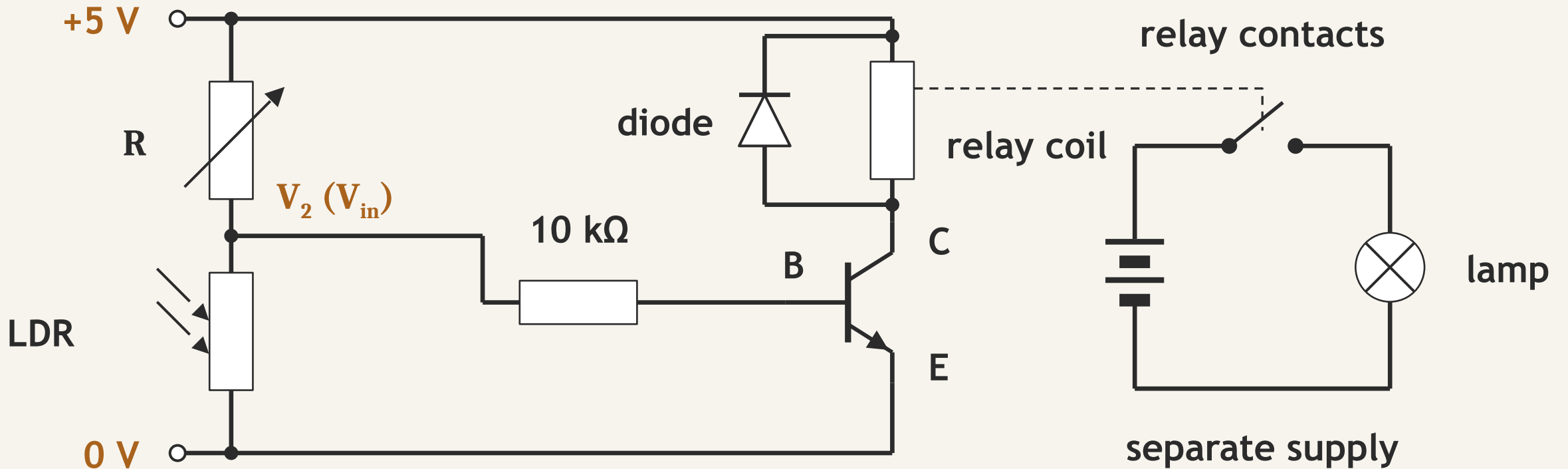
The pipe heater — one link, one mark

As the temperature falls...	Mark
the thermistor's resistance increases,	1
so V_2 (V_{in}) increases.	1
The transistor switches on,	1
so the relay's coil switches on and its contacts close,	1
and the heater switches on.	1

Leave out the resistance and you lose its mark.

Your turn – the porch light

Dark-sensing: on at night, through a relay



Describe what happens as the sun rises. Refer to the LDR, $V_2 (V_{in})$, the transistor, the relay and the lamp. (5 marks)

Swap and mark – the porch light

ANSWERS

One link, one mark

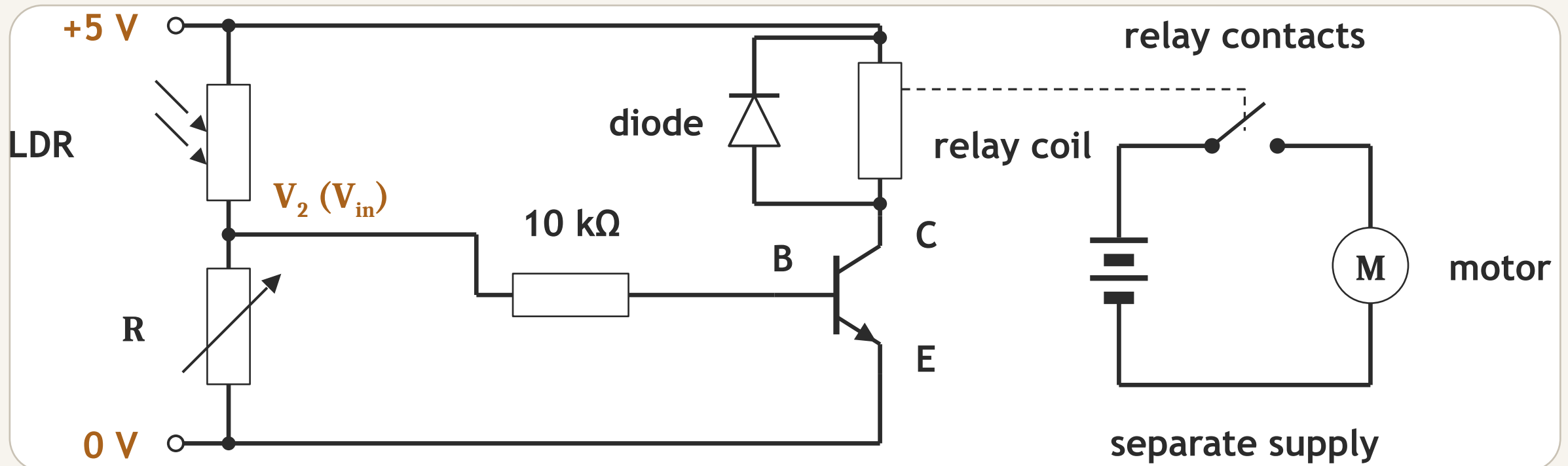
As the light level increases...	Mark
the LDR's resistance decreases,	1
so V_2 (V_{in}) decreases.	1
The transistor switches off,	1
so the relay's coil switches off and its contacts open,	1
and the lamp switches off.	1

Give each mark only for its own link, in this order.

Practical 4 – The Relay and the Protection Diode

Practical sheet on Teams · success criteria 5.3, 5.4 and 5.5

A relay switching a motor on its own supply. Yenka only.



Practical 5 – Find and Fix

Practical sheet on Teams · success criteria 2.6, 4.3-4.5 and 5.1-5.6

Test a circuit the way an engineer does.
Change one thing, then test again.
Not every test finds a fault.

planned test



expected result



initial test result



planned amendment



re-test result

Daily Review

Part 2 — Relays and the Protection Diode

- 1 What does a relay let a small circuit do?
switch a separate, bigger circuit
- 2 When does a relay coil make a voltage spike?
when it switches off
- 3 Where does the LDR go in a dark-sensing circuit?
at the bottom, as R_2
- 4 A changeover relay switches on. What do its two outputs do?
one goes off, one comes on
- 5 What does a description start with?
the sensor's resistance

PART 3

Putting It All Together

Ten questions from the booklet, then two from real exam papers.

By the end of this part...

- 1 I can interpret a control circuit using a divider, transistor, relay and output
- 2 I can describe the function of a relay and a protection diode

Your turn – the practice questions

Booklet p.29-30 – PRACTICE: Transistor Switching and Relays

Answer questions 1 to 10 in your booklet.
The marks are printed beside each question.
Questions 7 and 9 want the full description –
resistance, then V_2 (V_{in}), then the transistor.

Booklet p.29-30 – PRACTICE, Q1-10

PRACTICE – Answers, Q1-3

ANSWERS

Booklet p.29-30 – Transistor Switching and Relays

Q	The answer	Marks
1	a fast electronic switch – a small input voltage switches a much larger output current	1
2	base, collector and emitter	1
3	about 0.7 V	1

PRACTICE – Answers, Q4-5

ANSWERS

Booklet p.29-30 – Transistor Switching and Relays

Q	The answer	Marks
4	an electromagnetic switch – a small coil current closes a separate set of contacts	2
5	switching the coil off makes a voltage spike; the diode across it, in reverse, gives it a safe path	2

Q4 also wants: so a small circuit can switch a much larger one.

PRACTICE — Answers, Q6-8

ANSWERS

Booklet p.29-30 — Transistor Switching and Relays

Q	The answer	Marks
6	in the lower position, as R_2	1
7	as it darkens the LDR's resistance rises, so V_2 rises; at about 0.7 V the transistor switches the lamp on	3
8	the transistor circuit cannot carry a mains current; the relay's contacts switch it, and keep it separate	2

PRACTICE — Answers, Q9-10

ANSWERS

Booklet p.29-30 — Transistor Switching and Relays

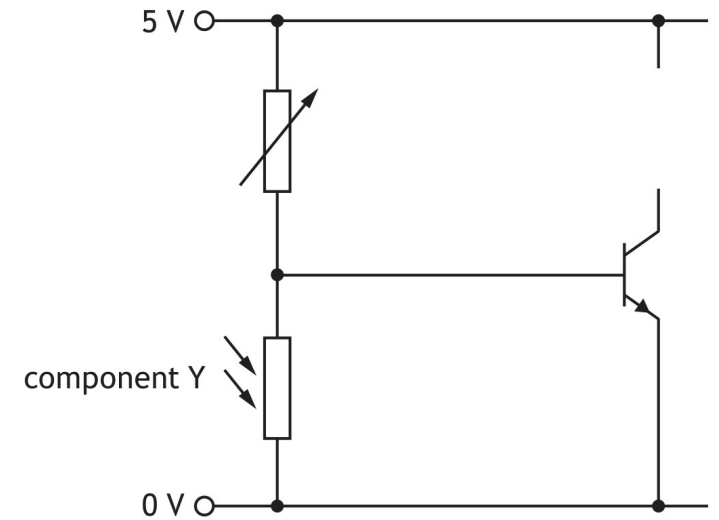
Q	The answer	Marks
9	thermistor in the upper position; as it heats, its resistance falls, so V_2 rises and the fan switches on	3
10	any two, with fixes — transistor legs reversed · diode missing or backwards · sensor in the wrong half	4

Q10 is 2 + 2: each mistake earns a mark, each fix earns another.

Exam practice

National 5 Engineering Science 2023 Q7 – a child's night light

Part	What it asks	Marks
7(a)	add the lamp symbol	1
7(b)	full name of component Y	1
7(c)	ammeter for base current	1
7(d)	what protects the transistor	1



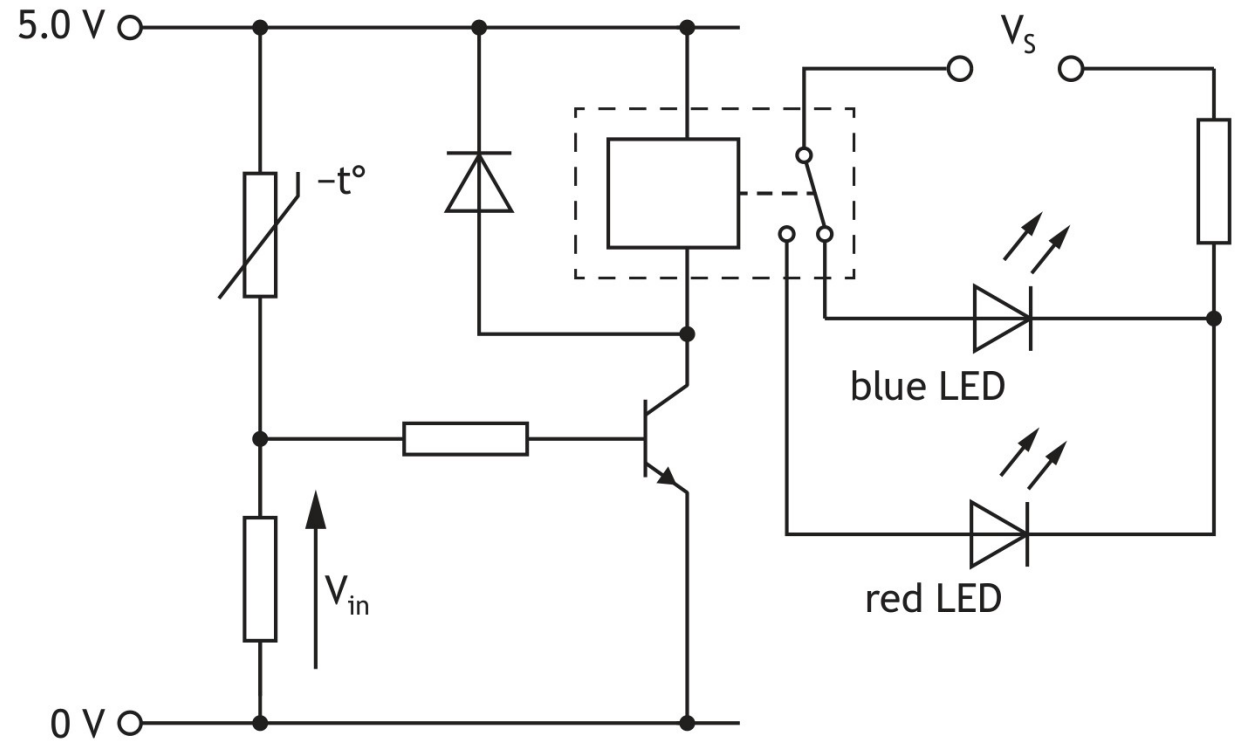
Exam practice – finish the kettle

National 5 Engineering Science 2024 Q10(d) – the kettle's LEDs

10(d) – 5 marks

Describe the circuit as the water heats up.

Where the marks are resistance, V_{in} , transistor or relay, and one for each LED



LOOKING AHEAD

Logic and Digital Control

Booklet 4a — when a signal is only ever on or off.

Plenary

- 1 What voltage switches an NPN transistor on?
about 0.7 V
- 2 Which connection goes to the output device?
the collector
- 3 What moves inside a relay when its coil switches on?
a metal arm – the contacts close
- 4 What makes the voltage spike in a relay circuit?
the coil, switching off
- 5 In a dark-sensing circuit, where is the LDR?
at the bottom, as R_2

Plenary

EXTRA PRACTICE

- 1 The base is at 0.5 V. Is the output device on?
no – below 0.7 V
- 2 Which connection goes to 0 V?
the emitter
- 3 Name the component that lets a 5 V circuit switch a 230 V fan.
a relay
- 4 What gives the coil's spike a safe path?
the protection diode
- 5 In a low-temperature circuit, where is the thermistor?
at the bottom, as R_2

Extension

- 1 A dark-sensing circuit switches on too early in the evening. What would you change, and why?
- 2 Why does a relay coil need a diode but a lamp does not?
- 3 The variable resistor is replaced by a fixed one. What does the circuit lose?

- 1 A frost heater comes on too late on a cold night. What would you change, and why?
- 2 Could you swap the LDR and the fixed resistor and get the same circuit back by turning R? Explain.
- 3 Why is the transistor drawn between the output device and 0 V, rather than between +V and the output device?

Course specification

N5 Engineering Science course specification v2.0, p.5

Analogue electronic control systems

- function of relays
- function of a protection diode in an electronic circuit
- explaining the switching function of a transistor
- operating an electronic control circuit, which includes a variable voltage divider, transistor, relay and output transducer